Topic 1.2 – Earth's Life Support Systems

"Just as human activity is upsetting the Earth's carbon cycle, our actions are altering the water cycle." – David Suzuki

Water and carbon support life on Earth, utilised by flora, fauna and humans. 71% of the Earth's surface is covered in water however 68% of the freshwater is locked in ice and glaciers. Water is moved and stored beneath our feet and this 30% is critically important to life on Earth.

Water and carbon are cycled between the land, oceans and atmosphere in open and closed systems, the processes within these cycles are inter-related.

Forests, soils, oceans and the atmosphere all store carbon and yet they are threatened and altered by human activity, this will be examined in detail through the Tropical Rainforest and the Arctic tundra case studies as well as at a global scale.

Physical changes in these cycles occur over time, from seconds to millions of years, and these changes can be seen at a range of scales, from individual plants or trees to vast ecosystems. With research and monitoring it is clear there is an increased need for global and national solutions to protect 'Earth's life support systems'.

1. How important are water and carbon to life on Earth?	
Key Ideas	**Content**
1.a. Water and carbon support life on Earth and move between the land, oceans and atmosphere.	• The importance of water in supporting life on the planet, the uses of water for humans, flora and fauna. • Carbon is the building block of life on Earth. It is available for use in the natural world and by humans. • Water and carbon cycling between the land, oceans and atmosphere through open and closed systems.
1.b. The carbon and water cycles are systems with inputs, outputs and stores.	• The distribution and size of the major stores in the carbon and water systems, including the atmosphere, oceans, water bodies, ice (cryosphere), soil, vegetation and groundwater. • The characteristics of the main inputs and outputs of the water cycle, including precipitation and snowmelt (ablation) and evapotranspiration. • The characteristics of the main inputs and outputs of the carbon cycle, including precipitation, photosynthesis, decomposition, weathering (including main forms of chemical weathering) respiration and combustion.
1.c. The carbon and water cycles have distinctive processes and pathways that operate within them.	• The processes of the water cycle, including evaporation, transpiration, condensation (including formation of clouds), precipitation (including causes of precipitation), interception, ablation, runoff (including overland flow and saturated overland flow), catchment hydrology (including infiltration, percolation, throughflow, groundwater flow and cryospheric processes). • The processes of the carbon cycle, including photosynthesis, respiration, decomposition, combustion (including natural and fossil fuel use), natural sequestration in oceans, vegetation, sediments and weathering.

<table>
<tr><th colspan="2">2. How do the water and carbon cycles operate in contrasting locations?</th></tr>
<tr><th>Key Ideas</th><th>Content</th></tr>
<tr><td>2.a. It is possible to identify the physical and human factors that affect the water and carbon cycles in a tropical rainforest.</td><td>

- **Case study** of a tropical rainforest, including:
  - water and carbon cycles specific to tropical rainforests, including the rates of flow and distinct stores. How an individual tree through to the rainforest as a whole can influence these cycles
  - physical factors affecting the flows and stores in the water cycle, including temperature, rock permeability and porosity and relief
  - physical factors affecting the flows and stores in the carbon cycle, including temperature, vegetation, organic matter in soil and the mineral composition of rocks
  - for one drainage basin in the tropical rainforest, explore the changes to the flows and stores within the water cycle caused by natural and human factors such as deforestation and farming factors
  - the impact of human activity, such as deforestation and farming, on carbon flows, soil and nutrient stores
  - strategies to manage the tropical rainforest such as afforestation and improved agriculture techniques that have positive effects on the water and carbon cycles.

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<tr><td>2.b. It is possible to identify the physical and human factors that affect the water and carbon cycles in an Arctic tundra area.</td><td>

- **Case study** of the Arctic tundra, including:
  - water and carbon cycles specific to Arctic tundra, including the rates of flow and distinct stores
  - physical factors affecting the flows and stores in the cycles, including temperature, rock permeability and porosity and relief
  - physical factors affecting the flows and stores in the carbon cycle, including temperature, vegetation, organic matter in soil and the mineral composition of rocks
  - seasonal changes in the water and carbon cycles in the Arctic tundra
  - the impact of the developing oil and gas industry on the water and carbon cycles
  - management strategies used to moderate the impacts of the oil and gas industry.

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3. How much change occurs over time in the water and carbon cycles?	
Key Ideas	**Content**
3.a. Human factors can disturb and enhance the natural processes and stores in the water and carbon cycles.	• Dynamic equilibrium in the cycles and the balance between the stores and the flows. • Land use changes, such as growth in urban areas, farming and forestry, as a catalyst for altering the flows and stores in these cycles. • How water extraction, including surface extraction and sub-surface groundwater extraction (including aquifers and artesian basins) impact the flows and stores in these cycles. • The impact of fossil fuel combustion and carbon sequestration on flows and stores of carbon. • Positive and negative feedback loops within and between the water and carbon cycles.
3.b. The pathways and processes which control the cycling of water and carbon vary over time.	• Short term changes to the cycles and the significance of these changes, including diurnal and seasonal changes of climate, temperature, sunlight and foliage. • Long term (millions of years) changes in the water and carbon cycles, including changes to stores and flows. • The importance of research and monitoring techniques to identify and record changes to the global water and carbon cycles; reasons why this data is gathered.

4. To what extent are the water and carbon cycles linked?	
Key Ideas	**Content**
4.a. The two cycles are linked and interdependent.	• The ways in which the two cycles link and are interdependent via oceans, atmosphere, cryosphere and vegetation. • How human activities cause changes in the availability of water and carbon (including fossil and terrestrial) stores, such as the use of these as resources. • The impact of long-term climate change on the water and carbon cycles.
4.b. The global implications of water and carbon management.	• Global management strategies to protect the carbon cycle as regulator of the Earth's climate, including afforestation, wetland restoration, improving agricultural practices and reducing emissions (including carbon trading and international agreements). • Global management strategies to protect the water cycle including improving forestry techniques, water allocations for domestic, industrial and agricultural use and drainage basin planning (including run-off, surface stores and groundwater).

5. Topic-specific skills:

- climate graphs
- simple mass balance
- rates of flow
- unit conversions
- analysis and presentation of field data.